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Title: Artificial Intelligence in Auditing: Revolutionizing Compliance and Assurance Processes

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Summary:

This paper investigates the transformative role of artificial intelligence (AI) in revolutionizing compliance and assurance processes within financial auditing. The authors argue that AI technologies, such as machine learning, natural language processing, and robotic process automation, are reshaping traditional auditing by automating routine tasks, enhancing compliance monitoring, and improving the reliability of assurance processes. The study highlights how AI can process vast volumes of financial data, ensure regulatory adherence, and provide real-time insights, thereby addressing the limitations of manual auditing methods in an increasingly complex financial landscape.

The methodology integrates a quantitative analysis of AI implementation across 30 audit engagements (2019-2023) with qualitative insights from interviews with audit professionals at mid-sized and large firms. Key findings show that AI tools reduce compliance verification time by 28% and improve error detection rates by 18% compared to traditional methods. A case study of a financial services firm demonstrates how AI-driven text analysis of contracts ensured compliance with IFRS standards, identifying discrepancies in 12% of sampled documents. Challenges include high initial costs, data integration issues with legacy systems, and the need for transparent AI models to meet regulatory scrutiny.

The paper concludes that AI enhances audit quality by enabling continuous monitoring and predictive compliance checks, but it emphasizes the importance of auditor training and ethical AI governance to mitigate risks like algorithmic bias. The authors advocate for standardized AI frameworks to facilitate adoption and call for further research into long-term impacts on audit firm structures. This work positions AI as a critical enabler for modern auditing, balancing efficiency gains with the need for human oversight to maintain trust in financial reporting.